

Core 4 — Completion Plan & Repository Blueprint

In-Memory Acceleration of Monte Carlo and ML-Based Financial Workloads
Status as of Sep 11, 2026 • Final submission due Sep 19, 2026 (demo video only)

1. Where Things Stand Right Now

Component	Contributor	Status
Monte Carlo pricer + dataset	Nini	Done — 10,000-scenario dataset generated, validated against Black-Scholes (~0.12% error)
Neural-network surrogate	Rupsaa	In progress — training pipeline should be running on the real dataset per schedule
IMC mapping	Raya	In progress — architecture mapping / preliminary estimates due at this stage
Benchmarking & comparison	Ishika	Schema + starter scripts should exist; real numbers arrive as MC/NN/IMC results land

Reminder: the only required final deliverable is a single demo video — no code repo or written report is graded. The repo below is for your own reproducibility and team coordination, not a submission requirement.

The day-by-day plan continues on the next (landscape) page.

2. Milestone Plan (Today Onward)

Tasks below are grouped by milestone date, not locked to one person — pick up whatever's open based on who has time that day. The only hard requirement is that each milestone's tasks are done by that date.

Date	Open Tasks	Milestone
Sep 11 (today)	Confirm MC dataset is final and shared • Start/continue NN training on real dataset • Map the frozen NN architecture for IMC • Stand up benchmark harness	
Sep 12	Validate MC across more scenarios • NN accuracy tuning (log MAE/RMSE/R-squared) • Preliminary IMC latency/energy estimates • First MC-vs-NN runtime comparison plots	
Sep 13	Whoever's free: integrate and debug the full pipeline together	END-TO-END PIPELINE WORKS
Sep 14	Finalize/expand dataset if needed • Final NN training run • Final IMC analysis pass • Begin systematic benchmarks across all three methods	
Sep 15	Runtime/validation experiments • Accuracy + inference-latency experiments • IMC experiments (vary assumptions if time allows) • Build comparison plots	
Sep 16	Sanity-check financial validity • Finalize NN numbers • Finalize IMC assumptions in writing • Final tables/figures	RESULTS FROZEN
Sep 17	Each component: write a 1–2 sentence plain-English explanation of what it does • Draft demo video script tying MC → NN → IMC together	
Sep 18	Whole team: record and review the demo together	VIDEO FINISHED
Sep 19	Final check by everyone	SUBMIT

Bold milestones: **Sep 13** pipeline must run end-to-end • **Sep 16** results are frozen (no new features after this unless something is broken) • **Sep 18** video is done, not just started.

3. Recommended Git Repository Structure

One shared repo (or one repo per component, linked from a root README) with this layout:

```
core4-project/
|-- README.md          <- pipeline overview, team roles, how it all fits together
|-- requirements.txt    <- numpy, pandas, torch/tensorflow, scikit-learn, matplotlib
|-- .gitignore         <- __pycache__/, .ipynb_checkpoints/, venv/, large intermediate files
|-- docs/
|   |-- project_plan.pdf
|   |-- one_page_proposal.pdf
|   |-- core4_completion_plan.pdf (this document)
|-- monte_carlo/        <- primarily Nini, open to all
|   |-- README.md       <- method, validation numbers, dataset schema
|   |-- monte_carlo.py   <- reusable pricer + dataset generator, with docstrings
|   |-- monte_carlo.ipynb <- interactive validation notebook
|   |-- core4_mc_dataset_10000.csv
|-- neural_network/     <- primarily Rupsaa, open to all
|   |-- README.md       <- architecture, training setup, final metrics
|   |-- train.py / train.ipynb
|   |-- model.py        <- NN class definition
|   |-- saved_model/    <- final weights (or a download link if too large)
|-- imc_mapping/        <- primarily Raya, open to all
|   |-- README.md       <- modeling approach, all assumptions spelled out explicitly
|   |-- imc_estimate.py  <- MAC counts, data movement, latency/energy estimation
|-- benchmarking/       <- primarily Ishika, open to all
|   |-- README.md
|   |-- benchmark.py     <- shared results schema + runtime measurement
|   |-- compare.py       <- builds the MC vs NN vs NN+IMC comparison
|   |-- plots/          <- generated figures used in the demo video
|-- results/
|   |-- results_table.csv <- method, mae, rmse, r2, latency_ms, throughput,
|   |                     data_movement_bytes, energy_estimate
|   |-- final_figures/   <- frozen final plots (post Sep 16)
```

Why this shape

- Each component folder is self-contained with its own README — anyone can understand one piece without reading the whole repo.
- **results/** is the single source of truth for the numbers that go in the demo video — avoids four people quoting slightly different figures.
- **docs/** keeps the original planning documents alongside the code, so decisions are traceable later.
- A root README should contain the 3-sentence pipeline story (MC → NN → IMC) so anyone opening the repo cold understands the point in 10 seconds.

Each Component README Should Contain

- What it does, in 2–3 sentences
- The exact method/formula used (not just a link to theory)
- Actual validation or accuracy numbers achieved — not placeholders

- How to run it (exact commands)
- What it hands off downstream, and in what format

Generated to accompany the Core 4 hackathon project plan, proposal, and materials checklist.